

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Change]	263015	74.164777
1	[Java, ByteCode, Method, Member]	13529	3.814898
2	[Java, ByteCode, Parameter]	13293	3.748350
3	[Git, Commit]	12966	3.656143
4	[File, Git]	10911	3.076676
5	[Java, ByteCode, ParameterizedType, Bound]	7279	2.052527
6	[Java, ByteCode, Bound]	7240	2.041530
7	[Java, ByteCode, Member, Field]	3604	1.016253
8	[Java, ByteCode, Bound, WildcardType]	2947	0.830993
9	[Java, Value, ByteCode, Annotation]	2931	0.826481
10	[Xml, Element]	2162	0.609639
11	[Java, ByteCode, Constructor, Method, Member]	2128	0.600052
12	[Xml, Text]	1450	0.408870
13	[Java, ByteCode, Bound, TypeVariable]	1111	0.313279
14	[Java, ByteCode, Method, Member, Lambda]	968	0.272956
15	[Type, File, Java, ByteCode, ResolvedDuplicate...	888	0.250398
16	[Type, File, Java, Class, ByteCode]	846	0.238554
17	[Json, Key]	702	0.197949
18	[Value, Json, Scalar]	685	0.193156
19	[Java, Value, ByteCode, Primitive]	672	0.189490
20	[Type, File, Java, ByteCode, JavaType]	649	0.183005
21	[Java, ByteCode, Method, Member, GenericDeclar...	578	0.162984
22	[Type, File, Java, ByteCode, ExternalType]	411	0.115893
23	[Author, Git, Person]	299	0.084312
24	[Value, Array]	287	0.080928
25	[Committer, Git, Person]	252	0.071059
26	[Type, File, Java, Class, ByteCode, GenericDec...	237	0.066829
27	[Value, Property]	211	0.059498
28	[Type, File, Java, ByteCode, Interface]	190	0.053576
29	[Java, Value, Class, ByteCode]	189	0.053294

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

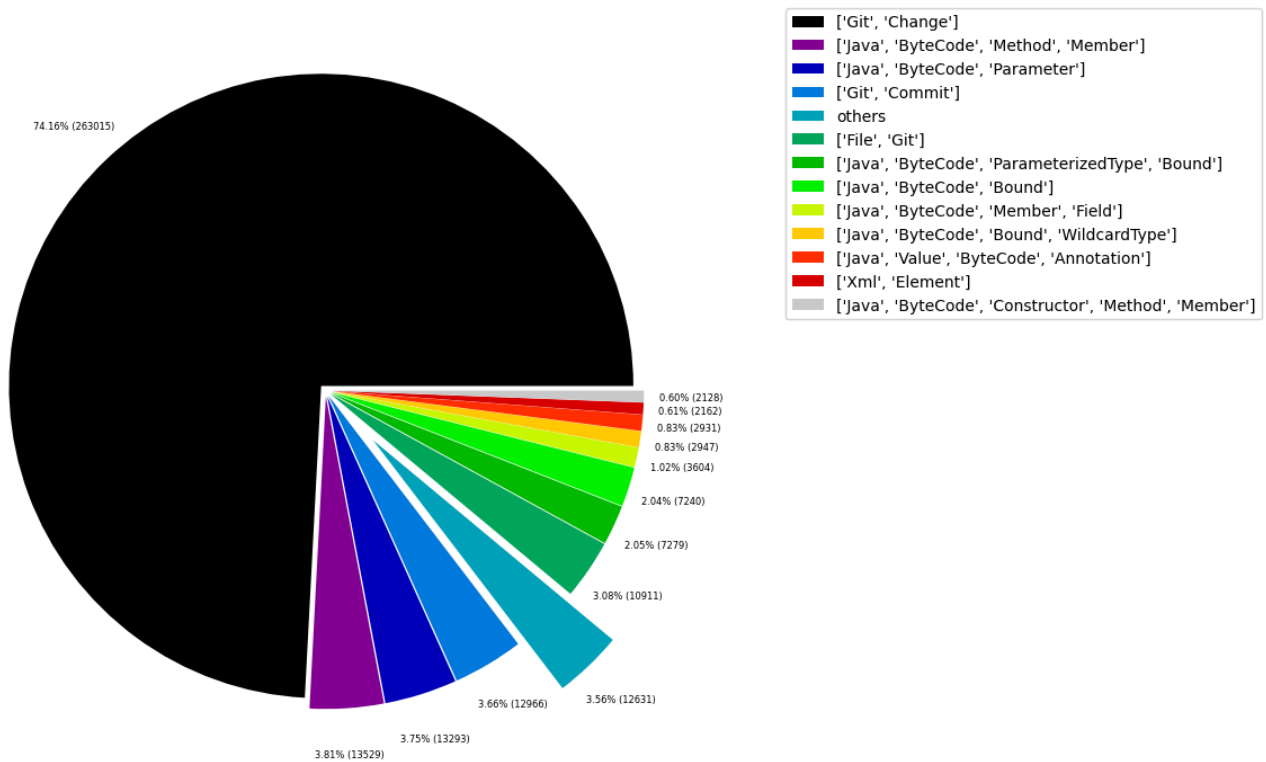


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000282
1	[Package, File, Json, NPM]	1	0.000282
2	[Repository, File, Git]	1	0.000282
3	[File, TS, Scan]	1	0.000282
4	[File, Json]	2	0.000564
5	[File]	3	0.000846
6	[Java, ByteCode, Constructor, Method, Member, ...]	4	0.001128
7	[Maven, Exclusion]	5	0.001410
8	[Value, Array, Json]	6	0.001692
9	[Dependency, NPM]	7	0.001974
10	[File, Maven, Xml, Pom, Document]	9	0.002538
11	[Type, File, Java, ByteCode, Void]	9	0.002538
12	[Java, ManifestSection]	9	0.002538
13	[File, Java, Manifest]	9	0.002538
14	[Artifact, File, Jar, Archive, Zip, Java]	9	0.002538
15	[File, Java, ServiceLoader]	10	0.002820
16	[File, Java, Properties]	12	0.003384
17	[Maven, PluginExecution]	16	0.004512
18	[Maven, ExecutionGoal]	16	0.004512
19	[Xml, Attribute]	18	0.005076
20	[jQAssistant, Rule, Concept]	19	0.005358
21	[Maven, Plugin]	21	0.005922
22	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.005922
23	[Maven, Configuration]	21	0.005922
24	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.006204
25	[Type, File, Java, ByteCode, Enum]	28	0.007895
26	[Type, File, Java, ByteCode, PrimitiveType]	30	0.008459
27	[Xml, Namespace]	36	0.010151
28	[Type, File, Java, ByteCode, Annotation]	44	0.012407
29	[Git, Branch]	45	0.012689

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

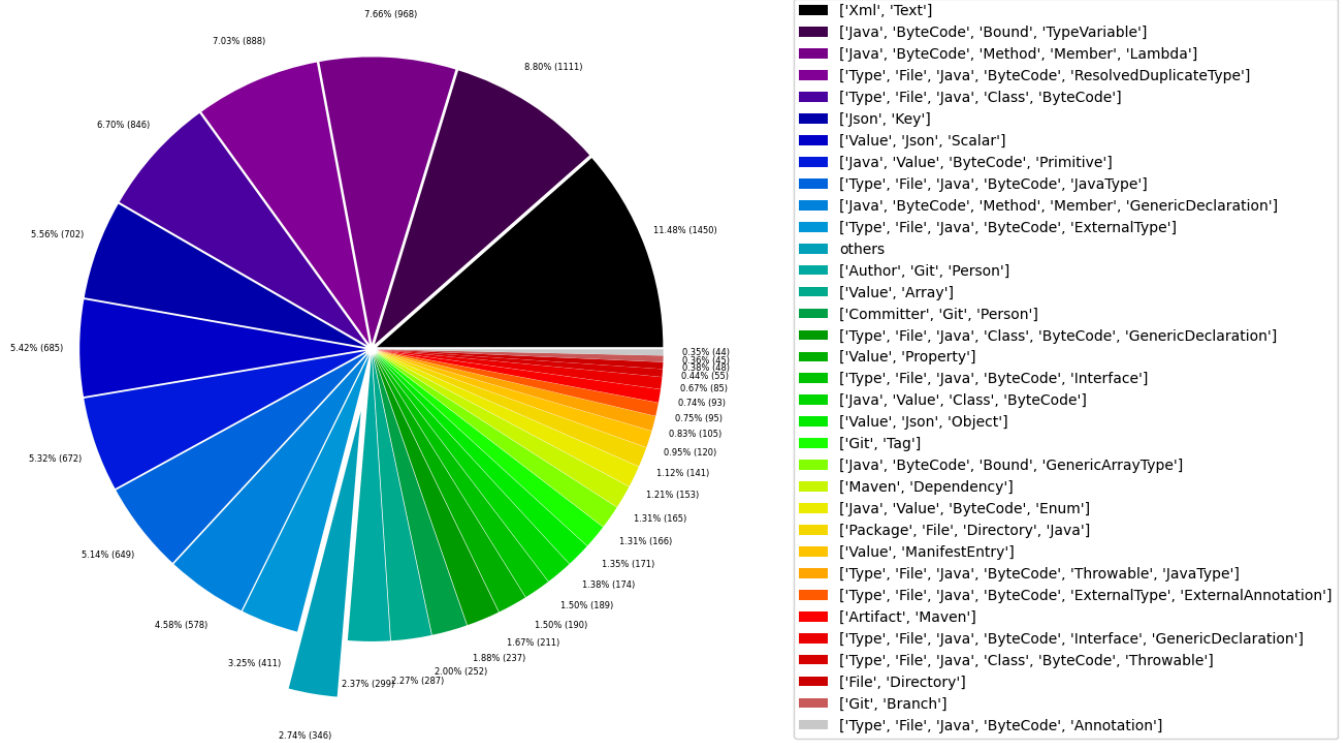


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

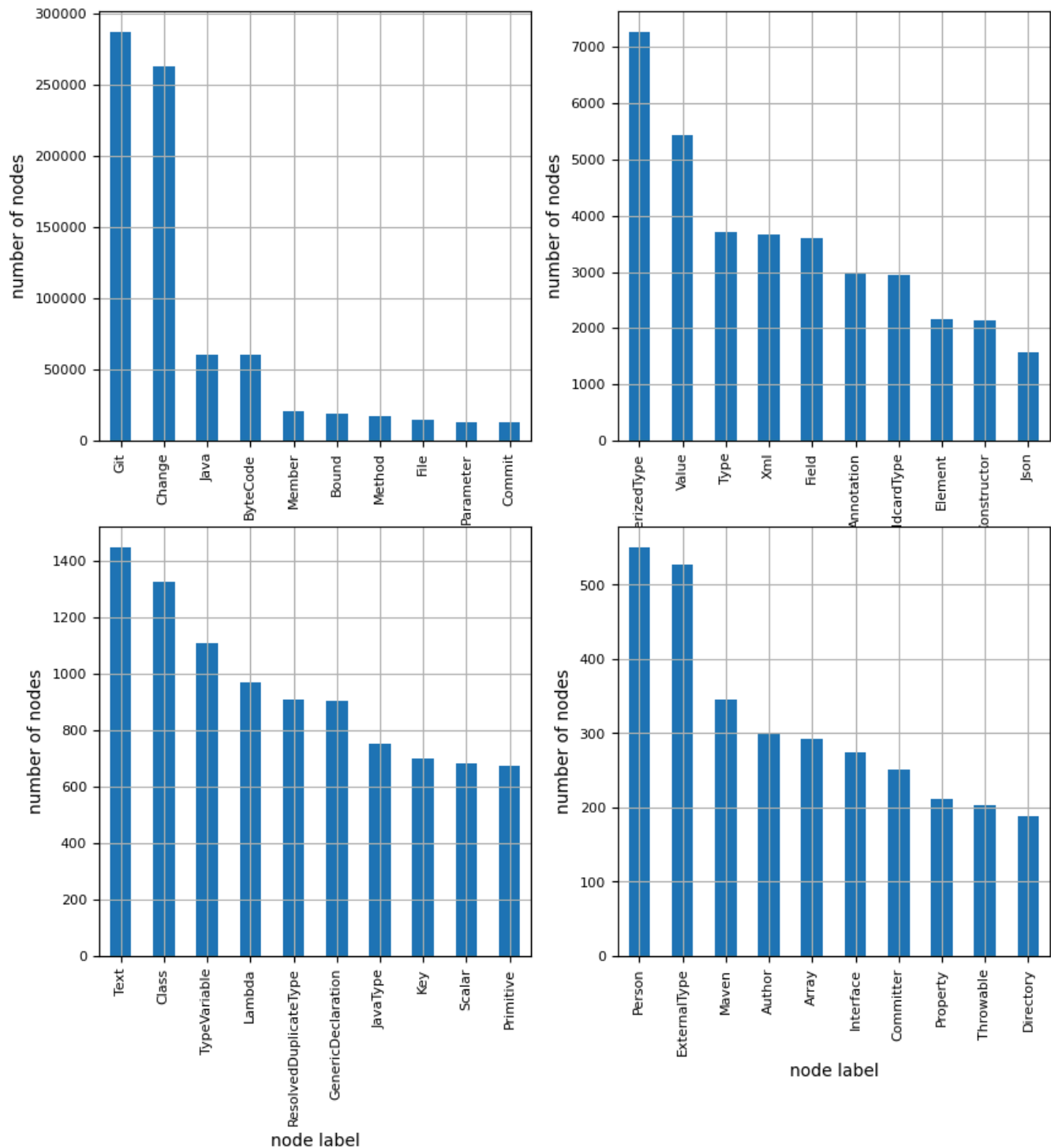
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	287660	81.114156
1	Change	263015	74.164777
2	Java	60697	17.115296
3	ByteCode	60507	17.061720
4	Member	20811	5.868271
5	Bound	18743	5.285137
6	Method	17207	4.852017
7	File	14872	4.193596
8	Parameter	13293	3.748350
9	Commit	12966	3.656143
10	ParameterizedType	7279	2.052527
11	Value	5428	1.530583
12	Type	3715	1.047553
13	Xml	3675	1.036274
14	Field	3604	1.016253
15	Annotation	2975	0.838888
16	WildcardType	2947	0.830993
17	Element	2162	0.609639
18	Constructor	2132	0.601180
19	Json	1570	0.442707
20	Text	1450	0.408870
21	Class	1327	0.374186
22	TypeVariable	1111	0.313279
23	Lambda	968	0.272956
24	ResolvedDuplicateType	910	0.256601
25	GenericDeclaration	904	0.254909
26	JavaType	754	0.212612
27	Key	702	0.197949
28	Scalar	685	0.193156
29	Primitive	672	0.189490
30	Person	551	0.155371
31	ExternalType	527	0.148603
32	Maven	346	0.097565
33	Author	299	0.084312
34	Array	293	0.082620
35	Interface	275	0.077544
36	Committer	252	0.071059
37	Property	211	0.059498
38	Throwable	203	0.057242
39	Directory	189	0.053294

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count` .

Total number of relationships: 1100439

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	263015	23.900916
1	MODIFIES	263015	23.900916
2	UPDATES	173566	15.772433
3	CREATES	61961	5.630571
4	DELETES	38056	3.458256
5	INVOKES	36595	3.325491
6	COMMITTED	25932	2.356514
7	DEPENDS_ON	22492	2.043912
8	OF_TYPE	21920	1.991932
9	DECLARES	21272	1.933047
10	OF_RAW_TYPE	17353	1.576916
11	HAS_PARENT	15699	1.426612
12	HAS	14429	1.311204
13	HAS_COMMIT	12966	1.178257
14	RETURNS	12844	1.167171
15	HAS_FILE	10911	0.991513
16	RENAMES	10568	0.960344
17	READS	9378	0.852205
18	HAS_ACTUAL_TYPE_ARGUMENT	8407	0.763968
19	HAS_NEW_NAME	6244	0.567410
20	OF_GENERIC_TYPE	5996	0.544873
21	RESOLVES_TO	5279	0.479718
22	SIMILAR	4078	0.370579
23	WRITES	3944	0.358402
24	CONTAINS	3936	0.357675
25	RETURNS_GENERIC	3593	0.326506
26	ANNOTATED_BY	2919	0.265258
27	REQUIRES	2230	0.202646
28	HAS_FIRST_CHILD	2162	0.196467
29	HAS_LAST_CHILD	2162	0.196467

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

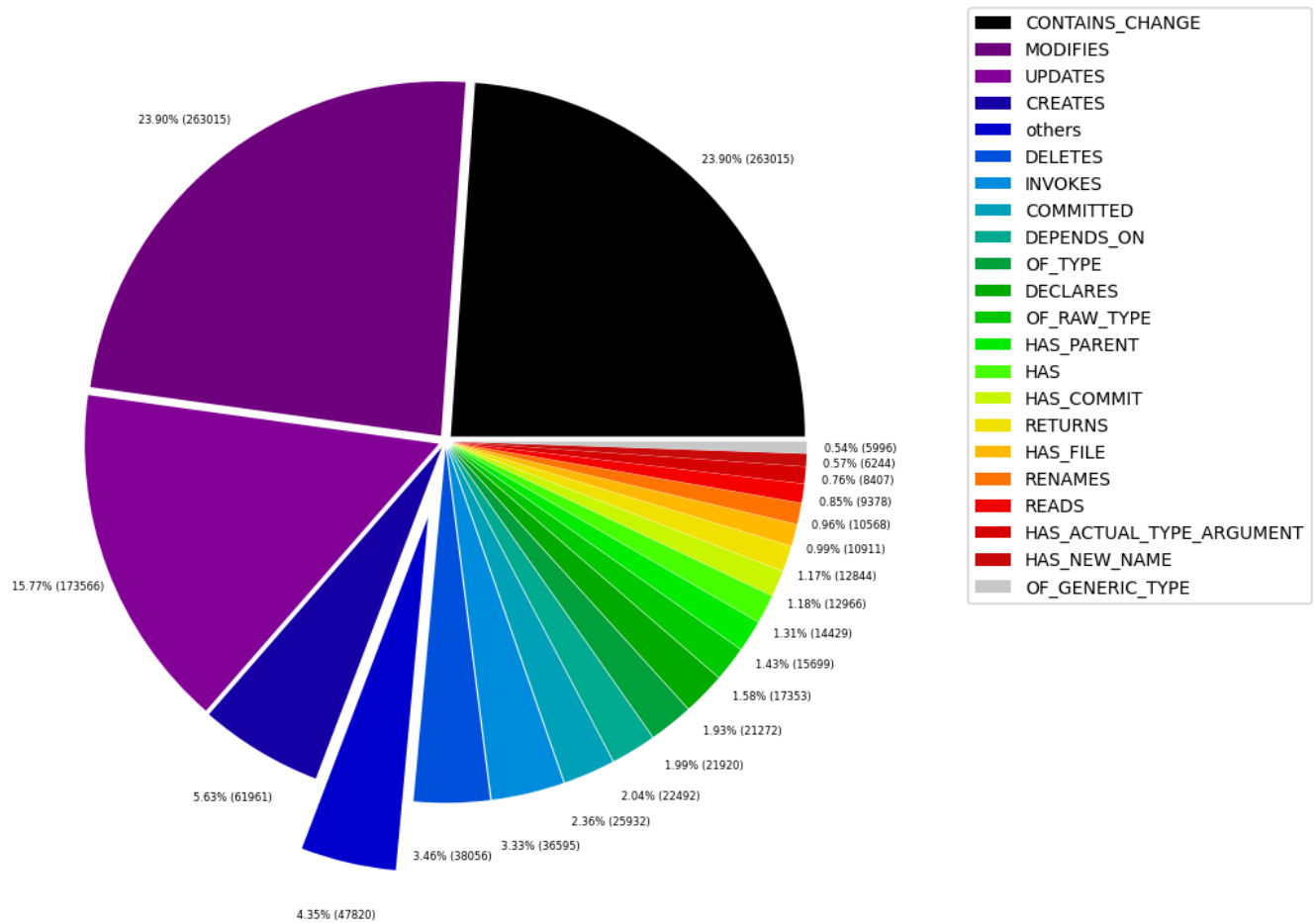


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

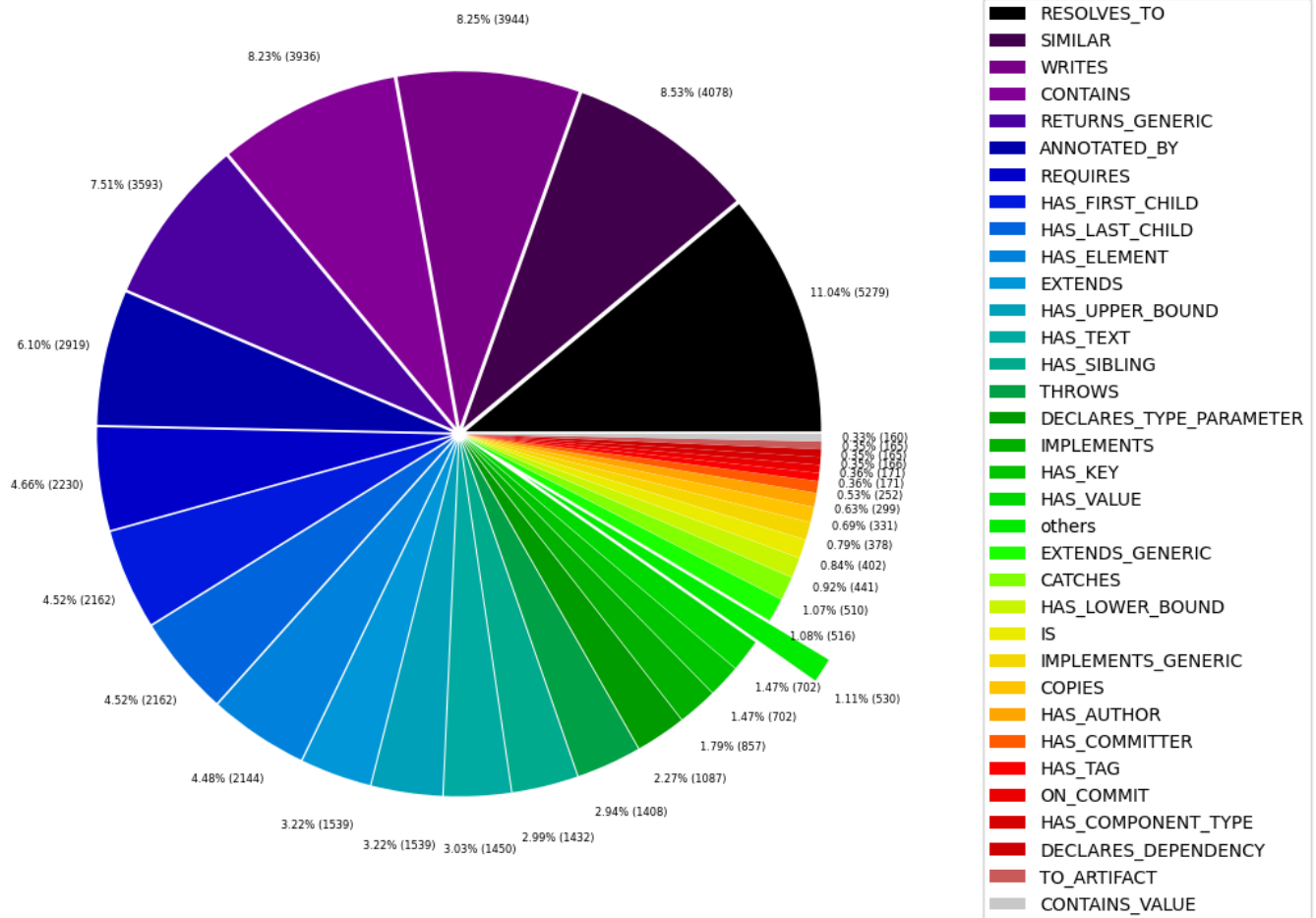
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000091
1	THROWS_GENERIC	5	0.000454
2	EXCLUDES	5	0.000454
3	DECLARES_DEV_DEPENDENCY	7	0.000636
4	DESCRIBES	9	0.000818
5	HAS_ROOT_ELEMENT	9	0.000818
6	HAS_GOAL	16	0.001454
7	HAS_EXECUTION	16	0.001454
8	OF_NAMESPACE	18	0.001636
9	HAS_ATTRIBUTE	18	0.001636
10	INCLUDES_CONCEPT	19	0.001727
11	IS_ARTIFACT	21	0.001908
12	HAS_CONFIGURATION	21	0.001908
13	USES_PLUGIN	21	0.001908
14	REQUIRES_TYPE_PARAMETER	24	0.002181
15	REQUIRES_CONCEPT	28	0.002544
16	DECLARES_NAMESPACE	36	0.003271
17	HAS_DEFAULT	39	0.003544
18	HAS_BRANCH	45	0.004089
19	HAS_HEAD	46	0.004180
20	COPY_OF	126	0.011450
21	CONTAINS_VALUE	160	0.014540
22	DECLARES_DEPENDENCY	165	0.014994
23	TO_ARTIFACT	165	0.014994
24	HAS_COMPONENT_TYPE	166	0.015085
25	ON_COMMIT	171	0.015539
26	HAS_TAG	171	0.015539
27	HAS_COMMITTER	252	0.022900
28	HAS_AUTHOR	299	0.027171
29	COPIES	331	0.030079

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Change]	263015	12966	
1	[Git, Change]	MODIFIES	[File, Git]	263015	263015	
2	[Git, Change]	UPDATES	[File, Git]	173566	263015	
3	[Git, Change]	CREATES	[File, Git]	61961	263015	
4	[Git, Change]	DELETES	[File, Git]	38056	263015	
5	[Java, ByteCode, Method, Member]	INVOKES	[Java, ByteCode, Method, Member]	22076	13427	
6	[Git, Commit]	HAS_PARENT	[Git, Commit]	15690	12966	
7	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	12966	1	
8	[Author, Git, Person]	COMMITTED	[Git, Commit]	12966	299	
9	[Committer, Git, Person]	COMMITTED	[Git, Commit]	12966	252	
10	[Repository, File, Git]	HAS_FILE	[File, Git]	10911	1	
11	[Git, Change]	RENAMES	[File, Git]	10568	263015	
12	[Java, ByteCode, Method, Member]	HAS	[Java, ByteCode, Parameter]	8476	13427	
13	[Java, ByteCode, Method, Member]	READS	[Java, ByteCode, Member, Field]	8448	13427	
14	[File, Git]	HAS_NEW_NAME	[File, Git]	6244	10911	
15	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6156	13293	
16	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3465	7240	
17	[Java, ByteCode, ParameterizedType, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3110	7279	
18	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2947	7279	
19	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, ParameterizedType, Bound]	2695	13293	
20	[Java, ByteCode, Constructor, Method, Member]	WRITES	[Java, ByteCode, Member, Field]	2519	2128	
21	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, TypeVariable]	2474	7279	
22	[Java, Value, ByteCode, Annotation]	OF_TYPE	[Type, File, Java, ByteCode, ExternalType, Ext...	2445	2931	
23	[Java, ByteCode, Constructor, Method, Member]	INVOKES	[Java, ByteCode, Constructor, Method, Member]	2195	2128	
24	[Xml, Element]	HAS_ELEMENT	[Xml, Element]	2144	2162	
25	[Java, ByteCode, Method, Member]	INVOKES	[Java, ByteCode, Constructor, Method, Member]	2121	13427	
26	[Java, ByteCode, Method, Member]	RETURNS	[Type, File, Java, ByteCode, JavaType]	2107	13427	
27	[Java, ByteCode, Constructor, Method, Member]	HAS	[Java, ByteCode, Parameter]	2075	2128	
28	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, Bound]	2061	13293	
29	[Java, ByteCode, Method, Member, Lambda]	INVOKES	[Java, ByteCode, Method, Member]	1933	968	

Graph Density

```
total_number_of_nodes (vertices): 354636
total_number_of_relationships (edges): 1100439
-> total directed graph density: 8.74986915199027e-06
-> total directed graph density in percent: 0.000874986915199027
```